

Delhi Public School

Subject: Physics | Grade: 12th | Class: 12th

Time: 1 Hour

Maximum Marks: 30

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

Section A

Answer all questions in this section. Questions 1 to 5 are Multiple Choice Questions, questions 6 to 10 are True/False questions, and questions 11 to 15 are Fill in the Blanks. Each question carries 1 mark.

Q1. Two point charges placed at a distance 'r' in air exert a force 'F' on each other. If the distance between them is doubled, the new force acting between them will be: [Easy] [1]

- a. $F/2$
- b. $F/4$
- c. $2F$
- d. $4F$

Expected Answer: $F/4$

Q2. If the potential difference applied across a given copper conductor is doubled, the drift velocity of free electrons inside the conductor will: [Moderate] [1]

- a. remain unchanged
- b. be halved
- c. be doubled
- d. become four times

Expected Answer: be doubled

Q3. Lenz's law of electromagnetic induction is a direct consequence of which of the following conservation laws? [Moderate] [1]

- a. Conservation of charge
- b. Conservation of mass
- c. Conservation of energy
- d. Conservation of linear momentum

Expected Answer: Conservation of energy

Q4. In a series LCR alternating current circuit, at resonance, the phase difference between the current and the source voltage is: [Hard] [1]

- a. 0 radians
- b. $\pi/2$ radians
- c. $\pi/4$ radians
- d. π radians

Expected Answer: 0 radians

Q5. The de Broglie wavelength of an electron accelerated through a potential difference of V volts is given by the expression: [Moderate] [1]

- a. $1.227 / \sqrt{V}$ nm
- b. $12.27 / \sqrt{V}$ nm
- c. $0.1227 / \sqrt{V}$ nm
- d. $122.7 / \sqrt{V}$ nm

Expected Answer: $1.227 / \sqrt{V}$ nm

Q6. The electrostatic potential inside a charged hollow spherical conductor is zero. [Easy] [1]

Expected Answer: False

Q7. A magnetic dipole placed in a uniform magnetic field always experiences a net non-zero translational force. [Moderate] [1]

Expected Answer: False

Q8. Self-inductance of a coil is also referred to as the electromagnetic inertia of the electrical circuit. [Easy] [1]

Expected Answer: True

Q9. When a light wave travels from an optically denser medium to an optically rarer medium, its frequency decreases. [Moderate] [1]

Expected Answer: False

Q10. The binding energy per nucleon is maximum for the iron nucleus (Fe-56), making it highly stable. [Hard] [1]

Expected Answer: True

Q11. The SI unit of electric flux is _____. [Easy] [1]

Expected Answer: Volt-meter

Q12. The magnetic susceptibility of a diamagnetic substance is always _____. [Moderate] [1]

Expected Answer: negative

Q13. A _____ is an electrical device used to step-up or step-down the alternating voltage based on mutual induction. [Easy] [1]

Expected Answer: transformer

Q14. The focal length of a plane mirror is _____. [Moderate] [1]

Expected Answer: infinite

Q15. In a p-n junction diode, the width of the depletion region _____ when a reverse bias voltage is applied. [Hard] [1]

Expected Answer: increases

Section B

Answer all short answer questions. Each question carries 2 marks.

Q16. State Gauss's Law in electrostatics and write down its mathematical expression for a charge enclosed in a closed surface. [Easy] [2]

Expected Answer: Gauss's law states that the total electric flux passing through a closed surface is equal to $1/\epsilon_0$ times the total net charge enclosed by the surface. Mathematically: $\oint \mathbf{E} \cdot d\mathbf{S} = q_{\text{enclosed}} / \epsilon_0$.

Q17. A wire of resistance R is stretched uniformly such that its length is doubled. Calculate the new resistance of the wire in terms of its original resistance. [Moderate] [2]

Expected Answer: When a wire is stretched to double its length ($l' = 2l$), its volume remains constant, meaning the area of cross-section becomes half ($A' = A/2$). Since $R = \rho l/A$, the new resistance $R' = \rho(2l)/(A/2) = 4 * (\rho l/A) = 4R$. Thus, the resistance becomes four times the original resistance.

Q18. Explain why the iron core of a transformer is laminated. What specific energy loss does this design minimize? [Hard] [2]

Expected Answer: The iron core of a transformer is laminated to break the continuous conducting paths for circulating currents. This design minimizes energy losses caused by Eddy Currents, which would otherwise generate heat and reduce transformer efficiency.

Section C

Answer all long answer questions. Each question carries 3 marks.

Q19. Derive an expression for the electric field intensity at a point on the axial line of an electric dipole of dipole moment 'p'. [Moderate] [3]

Expected Answer: Consider a dipole with charges $-q$ and $+q$ separated by distance $2a$. Let P be a point on the axis at distance r from the center. The electric fields due to $+q$ and $-q$ are: $E_+ = q / (4\pi\epsilon_0(r-a)^2)$ and $E_- = q / (4\pi\epsilon_0(r+a)^2)$. The net axial electric field $E = E_+ - E_- = [q / (4\pi\epsilon_0)] * [(4ra) / (r^2 - a^2)^2] = (2pr) / (4\pi\epsilon_0(r^2 - a^2)^2)$, where $p = 2qa$. For a short dipole where $r \gg a$, the field simplifies to $E = 2p / (4\pi\epsilon_0 r^3)$.

Q20. State the Biot-Savart Law in vector form. Use this law to derive the expression for the magnetic field at the center of a circular current-carrying loop of radius R . [Moderate] [3]

Expected Answer: Biot-Savart Law states $dB = (\mu_0/4\pi) * I(dl \times r) / r^3$. For a circular loop of radius R carrying current I , at the center, the angle between the length element dl and position vector r is 90 degrees. Thus, $dB = (\mu_0/4\pi) * (I dl \sin 90 / R^2) = (\mu_0/4\pi) * (I dl / R^2)$. Integrating around the loop: $B = \oint dB = (\mu_0 I / 4\pi R^2) * \oint dl$. Since $\oint dl = 2\pi R$, $B = (\mu_0 I / 4\pi R^2) * 2\pi R = \mu_0 I / 2R$.

Q21. With the help of a neat circuit diagram, explain the working of a junction diode as a full-wave rectifier. Sketch the input and output waveforms. [Hard] [3]

Expected Answer: A full-wave rectifier uses a center-tapped transformer and two junction diodes (D_1 and D_2). During the positive half cycle of AC input, terminal A is positive and B is negative, making D_1 forward-biased (conducts) and D_2 reverse-biased (non-conducting). Current flows through load resistor R_L . During the negative half cycle, A is negative and B is positive, making D_1 reverse-biased and D_2 forward-biased (conducts). Current flows through R_L in the same direction as before. Thus, both half cycles are rectified to produce a continuous, unidirectional pulsating DC output.

— End of Question Paper —